

HOPETOUN VIC, Australia

WMO: 948380

Lat: 35.7150S

Lon: 142.3569E

Elev: 256

StdP: 14.56

Time Zone: 10.00 (AUS)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.5	33.8	21.9	16.6	75.8	25.4	19.6	69.8	27.2	56.3	25.1	56.4	3.3	90	0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.4	103.5	66.4	98.9	65.5	94.9	64.5	71.1	86.9	69.2	87.3	67.6	86.0	14.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.1	100.7	74.5	64.5	91.8	73.4	61.8	83.2	72.7	35.1	86.2	33.4	86.9	32.1	86.3	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	22.9	20.3	27.2	111.2	1.9	3.0	25.8	113.3	24.7	115.1	23.7	116.8	22.4	118.9
			26.4	73.9	1.9	2.5	25.0	75.7	23.9	77.1	22.9	78.5	21.5	80.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	75.3	73.6	69.1	61.5	54.4	49.4	48.3	49.8	53.8	60.3	66.4	70.7	(6)
(7)		DBStd	11.50	7.98	7.55	7.64	6.39	4.82	3.80	3.97	4.57	5.82	7.42	8.36	8.27	(7)
(8)		HDD50	227	0	0	0	1	12	55	80	59	18	2	0	0	(8)
(9)		HDD65	2628	6	12	41	143	329	468	516	471	342	189	81	30	(9)
(10)		CDD50	4235	784	659	593	346	148	36	29	53	132	321	492	642	(10)
(11)		CDD65	1164	325	251	169	38	1	0	0	6	43	123	208	(11)	
(12)		CDH74	16773	4628	3295	2127	544	19	0	0	9	152	882	1935	3182	(12)
(13)		CDH80	8713	2664	1743	1018	176	0	0	0	39	372	981	1720	(13)	
(14)	Wind	WSAvg	9.9	11.3	11.0	10.0	8.7	8.7	8.0	9.1	9.4	9.8	10.5	11.0	11.3	(14)
(15)	Precipitation	PrecAvg	13.9	1.0	0.7	0.9	1.1	1.5	1.1	1.3	1.5	1.5	1.4	1.1	0.8	(15)
(16)		PrecMax	22.6	3.5	3.5	3.9	5.6	3.7	3.6	2.9	3.0	4.1	4.7	4.4	3.0	(16)
(17)		PrecMin	4.1	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.2	0.2	0.0	0.0	0.0	(17)
(18)		PrecStd	4.1	1.0	0.7	1.0	1.1	1.0	0.8	0.7	1.0	1.0	1.0	1.0	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	109.2	106.5	101.4	91.7	76.6	65.6	65.7	74.7	85.1	95.3	101.6	107.4	(19)
(20)			MCWB	67.9	67.5	66.0	63.1	58.3	55.3	53.4	55.2	57.9	61.1	64.3	66.3	(20)
(21)		2%	DB	104.8	99.5	95.4	84.7	71.9	62.2	61.7	68.4	78.4	89.2	95.6	101.1	(21)
(22)			MCWB	67.2	66.5	64.9	61.0	57.1	53.3	52.6	53.3	57.1	59.9	63.2	65.0	(22)
(23)		5%	DB	98.9	94.9	90.5	80.1	67.7	60.0	59.1	63.9	72.2	83.7	90.7	95.1	(23)
(24)			MCWB	66.2	65.9	64.1	59.1	55.7	52.6	51.6	51.9	54.9	58.0	61.9	63.7	(24)
(25)		10%	DB	93.8	90.6	85.4	75.6	64.5	57.9	56.8	60.4	67.6	78.2	85.7	89.9	(25)
(26)			MCWB	65.0	64.8	63.0	57.8	55.2	51.8	50.7	51.3	53.5	57.3	61.2	62.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.5	74.0	70.1	65.4	61.2	57.9	56.6	57.1	60.7	65.9	69.3	71.9	(27)
(28)			MCDWB	88.2	90.6	88.4	79.8	68.2	61.3	60.8	67.8	74.8	79.9	86.0	85.7	(28)
(29)		2%	WB	71.3	70.9	68.2	63.2	59.1	55.3	54.2	54.9	58.5	62.6	66.9	68.3	(29)
(30)			MCDWB	86.7	86.9	83.2	75.6	67.8	60.0	59.0	64.2	73.1	78.6	84.1	85.2	(30)
(31)		5%	WB	69.3	68.8	66.4	61.3	57.5	53.8	52.7	53.3	56.4	60.5	64.7	66.3	(31)
(32)			MCDWB	89.6	85.9	82.3	75.3	65.3	58.2	57.5	62.0	69.2	76.8	83.3	87.0	(32)
(33)		10%	WB	67.4	66.9	64.6	59.5	55.9	52.6	51.2	51.8	54.5	58.5	62.6	64.3	(33)
(34)			MCDWB	89.0	85.0	82.8	73.6	63.1	56.9	55.8	59.3	65.7	75.3	79.9	86.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	29.4	28.1	26.1	24.8	20.6	18.3	18.1	21.5	25.7	29.1	28.7	29.4	(35)
(36)			MCDBR	36.3	33.6	32.0	30.8	25.2	20.9	20.7	27.7	33.1	37.7	35.7	35.8	(36)
(37)			MCWBR	11.9	11.0	11.6	14.0	13.7	13.2	12.9	14.7	15.5	15.3	12.3	12.6	(37)
(38)		5% WB	MCDWR	27.9	25.5	25.6	24.9	20.5	17.9	18.1	23.4	28.1	29.5	28.7	27.1	(38)
(39)			MCWBR	11.6	9.4	10.8	12.1	11.8	12.2	12.2	13.9	14.4	13.5	11.3	11.7	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.340	0.335	0.318	0.308	0.300	0.290	0.290	0.301	0.322	0.332	0.340	0.331	(40)
(41)		taud		2.481	2.508	2.548	2.553	2.556	2.581	2.581	2.529	2.464	2.447	2.456	2.495	(41)
(42)		Ebn at Noon		316	310	304	289	273	268	274	286	297	308	313	320	(42)
(43)		Edh at Noon		37	34	30	27	23	21	23	27	33	36	37	36	(43)
(44)	All-Sky Solar Radiation	RadAvg		2486	2224	1798	1306	897	721	775	1079	1478	1945	2230	2466	(44)
(45)		RadStd		112	108	81	61	42	50	39	67	107	125	126	99	(45)

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	+220	+157	

Nomenclature: See separate page